Enhanced Grover's Algorithm with Adaptive Techniques for Multi-Target Database Search

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I. Abstract

Quantum computing has been considered a revolutionary paradigm shift in the field of computational science since it provides solutions for some infeasible problems for classical computers. One of the famous quantum algorithms which have been developed by Grover is the Grover algorithm, which is known to provide a quadratic speed up for unstructured database search problems. Nevertheless, one of its limitations is that it is mainly applicable in single target situations and requires a fixed number of iterations. These limitations limit the practicality of Grover algorithm in many cases where multiple targets can exist but their number is not known or variable. Therefore, in this paper, Enhanced Grover's Algorithm with Adaptive Techniques (EGAAT) will be developed to efficiently perform multi-target searches in databases. The methodology used in the new algorithm incorporates three major features: (1) a multi-target oracle design; (2) an adaptive iteration control using quantum amplitude estimation, and (3) adaptively scheduling diffusion. Simulation-based evaluation using Qiskit-style benchmark settings shows that EGAAT can achieve success probabilities above 94% for several multi-target scenarios across databases ranging from 64 to 4096 entries, with query complexity $\mathcal{O}(\sqrt{N/k})$. Comparative evaluation against standard Grover search, fixed-point Grover variants, and quantum walk-based search indicates that the proposed method improves the success-to-depth trade-off and is better aligned with near-term intermediate-scale quantum hardware constraints.

Index Terms - Quantum computing, Grover's algorithm, multi-target search, amplitude amplification, quantum amplitude estimation, adaptive oracle, adaptive diffusion, NISQ hardware, Qiskit.

II. Introduction

A. Problem Statement

The problem of searching an unstructured database for one or more target entries is fundamental in computer science and information technology. In classical computing, an algorithm performing unstructured search must examine, on average, $N/2$ entries in a database of size N, giving time complexity $\mathcal{O}(N)$. While classical optimizations such as hashing and binary search reduce complexity under structural assumptions, they do not generalize to genuinely unstructured or dynamically changing databases.

Grover's quantum search algorithm achieves the task in $\mathcal{O}(\sqrt{N})$ oracle queries through quantum superposition and interference [1]. This quadratic speedup has been proven optimal for unstructured search [2]. However, the original algorithm is fundamentally designed around a known marked subspace. When extended naively to multi-target settings, its optimal iteration count depends on the number of marked states k. If k is unknown, a fixed iteration strategy can produce amplitude over-rotation or under-rotation, significantly reducing the success probability.

This creates the central research problem addressed in this paper: how can Grover's algorithm be generalized for efficient and reliable multi-target search without requiring advance knowledge of the number of targets?

B. Proposed Solution

The proposed Enhanced Grover's Algorithm with Adaptive Techniques (EGAAT) addresses this limitation using a three-layer adaptive framework. First, the oracle layer is redesigned to mark an arbitrary set of target states within a single quantum search process. Second, a lightweight quantum amplitude estimation stage is used to estimate the number of targets k and dynamically compute the number of Grover iterations. Third, the diffusion operator is modified using an adaptive scaling schedule based on the estimated target density $\rho = k/N$.

C. Research Gap

Existing work on Grover search mostly focuses on single-target settings. Multi-target variants often assume that k is known or rely on restart strategies. Fixed-point Grover variants improve robustness but usually require deeper circuits and additional overhead [5]. Quantum walk search methods are powerful for structured search spaces but add graph-dependent implementation complexity [9], [10]. A unified framework that combines multi-target oracle design, dynamic iteration control, and adaptive diffusion scheduling while remaining hardware-conscious is still needed.

D. Motivation

Near-term intermediate-scale quantum (NISQ) processors are limited by qubit count, gate noise, connectivity, and coherence time [13]. Therefore, quantum search methods intended for practical deployment must minimize circuit depth and avoid unnecessary ancilla overhead. Multi-objective search

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techniques can be used directly for cryptography, genome pattern matching, constraint solving, and anomaly detection. A robust adaptive search method can bridge the gap between theoretical Grover speedup and practical quantum search applications.

E. Contributions

The main contributions of this paper are:

1) A multi-target oracle construction for marking arbitrary target states in a unified quantum circuit.
2) A dynamic iteration control strategy based on amplitude estimation to approximate k and compute an appropriate iteration count.
3) An adaptive diffusion scheduling strategy that changes amplification strength based on estimated target density.
4) A simulation-based evaluation against standard Grover search, fixed-point Grover search, and quantum walk search.
5) A noise-resilience discussion using depolarizing error settings relevant to NISQ devices.

III. Literature Review

A. Foundations of Quantum Search

Grover introduced the quantum search algorithm in 1996, showing that an unstructured database can be searched in $\mathcal{O}(\sqrt{N})$ queries [1]. The algorithm alternates between an oracle operator that flips the phase of marked states and a diffusion operator that performs inversion about the mean. Boyer *et al.* later established tight bounds for quantum searching and confirmed the optimality of Grover's quadratic speedup [2]. Bennett *et al.* analyzed the strengths and limitations of quantum computation under oracle models, further clarifying the role of quantum search [3].

B. Multi-Target and Fixed-Point Grover Search

Multiple target search is an extension of the marked subspace for multiple target states to k marked states. Quantum amplitude amplification and estimation were first proposed by Brassard et al., who have offered a method to estimate solution counts and amplify the marked subspaces [4]. Fixed point quantum search was proposed by Yoder et al., wherein over rotation is avoided; however, it requires deep circuits [5]. Grover et al. studied partial quantum search, finding that any changes in the diffusion steps can make the search more vulnerable [6].

C. Amplitude Estimation and Adaptive Strategies

Amplitude estimation using quantum computing refers to the process used to estimate probability amplitudes of the marked state. Quantum Amplitude Estimation Algorithms utilize quantum phase estimation algorithms, which may sometimes necessitate the creation of complex quantum circuits. A quantum amplitude estimation algorithm that does not use quantum phase estimation using maximum likelihood is provided by Suzuki et al. [7]. Quantum counting using approximate methods has been made simpler by Aaronson and Rall, stressing adaptation techniques [8].

D. Quantum Walk-Based Search

Quantum walk search provides an alternative to amplitude amplification. Shenvi, Kempe, and Whaley showed quantum-walk search on a hypercube [9]. Ambainis developed quantum walk algorithms for element distinctness [10], while Magniez *et al.* provided a broader framework for search via quantum walk [11]. These approaches are useful for structured problems but can be less direct for purely unstructured database search.

E. NISQ Hardware Considerations

Preskill described the NISQ era and emphasized the need for shallow circuits and noise-aware design [13]. Kandala et al., in [14], proposed variational quantum algorithms that could be applied efficiently on real-world hardware systems. These researchers also encouraged the development of hybrid techniques that require lesser depths. Gilliam et al., in [15], described the Grover Adaptive Search Algorithm for solving constrained binary optimization problems.

IV. Theoretical Foundations

A. Grover's Algorithm Review

Let $N = 2^n$ be the size of the search space and let k denote the number of marked states. The initial state is the uniform superposition

$$|\psi_0\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle. \tag{1}$$

Standard Grover search applies an oracle U_f and a diffusion operator D repeatedly:

$$|\psi_m\rangle = (DU_f)^m|\psi_0\rangle. \tag{2}$$

The optimal number of iterations for k marked states is approximately

$$m^*(k) = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{k}} \right\rfloor. \tag{3}$$

If k is unknown, selecting m incorrectly may reduce the probability of measuring a marked state.

B. Multi-Target Oracle

For a target set $S = \{s_1, s_2, \ldots, s_k\}$, the oracle applies

$$U_f|x\rangle = (-1)^{f(x)}|x\rangle, \tag{4}$$

where $f(x) = 1$ if $x \in S$ and $f(x) = 0$ otherwise. For each target s, the binary pattern is matched by applying NOT gates to zero-valued bit positions, applying a multi-controlled-Z operation, and restoring the NOT gates. This produces a phase flip for each selected target state.

Algorithm 1 EGAAT Multi-Target Search

Require: Search size $N = 2^n$, target oracle definition, estimation iterations $M = \{1, 2, 4, 8, 16\}$
Ensure: Measured marked state with high probability
1: Prepare $|\psi_0\rangle = H^{\otimes n}|0\rangle^{\otimes n}$
2: Construct multi-target oracle U_f
3: **for** each $m \in M$ **do**
4: Execute $(DU_f)^m|\psi_0\rangle$
5: Estimate marked probability p_m from measurement counts
6: **end for**
7: Fit p_m to $\sin^2((2m+1)\theta)$ and obtain $\hat{\theta}$
8: Compute $\hat{k} = N\sin^2(\hat{\theta})$ and $\hat{\rho} = \hat{k}/N$
9: Compute $m^* = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{\hat{k}}} \right\rfloor$
10: Compute $\alpha(\hat{\rho})$
11: Execute $(D_\alpha U_f)^{m^*}|\psi_0\rangle$
12: Measure and return the observed state

C. Amplitude Estimation for Unknown Target Count

EGAAT uses a maximum-likelihood amplitude estimation style procedure. Grover circuits are executed for iteration values

$$m \in \{1, 2, 4, 8, 16\}. \tag{5}$$

The measured marked-state probability p_m is fitted to

$$P_m(\theta) = \sin^2((2m+1)\theta), \tag{6}$$

using the loss

$$L(\theta) = \sum_m |p_m - \sin^2((2m+1)\theta)|^2. \tag{7}$$

The estimated target count is then

$$\hat{k} = N\sin^2(\hat{\theta}). \tag{8}$$

D. Adaptive Diffusion Scheduling

The standard diffusion operator is

$$D = 2|\psi_0\rangle\langle\psi_0| - I. \tag{9}$$

EGAAT uses a scaled diffusion operator

$$D_\alpha = (1-2\alpha)I + 2\alpha|\psi_0\rangle\langle\psi_0|, \tag{10}$$

where α depends on the estimated density $\rho = \hat{k}/N$:

$$\alpha(\rho) = \begin{cases} 1.0, & \rho < 0.2, \\ 0.45 + 0.55(0.5-\rho)/0.3, & 0.2 \le \rho < 0.5, \\ (1-\rho)\cdot 0.5, & \rho \ge 0.5. \end{cases} \tag{11}$$

This schedule reduces over-amplification in dense target regimes and preserves stronger amplification in sparse regimes.

V. Methodology

A. Overall EGAAT Pipeline

The EGAAT pipeline consists of oracle configuration, target-density estimation, adaptive iteration selection, diffusion scheduling, and final measurement. Algorithm 1 summarizes the process.

TABLE I
Direct Pipeline Verification from Implementation Output

N	k	$\hat{k}$	m^*	P_{meas}	Depth
32	4	4	2	0.9463	24
–	3	3	1	0.9492	14

B. Implementation Structure

The implementation can be organized into three core classes: `MultiTargetOracle`, `QAEEstimator`, and `AdaptiveDiffusion`. A simplified oracle construction is shown below.

Listing 1. Simplified multi-target oracle construction.

```
class MultiTargetOracle:
    def __init__(self, targets, num_qubits):
        self.targets = targets
        self.num_qubits = num_qubits
        self.circuit = QuantumCircuit(num_qubits)
        self._build_oracle()

    def _build_oracle(self):
        for target in self.targets:
            binary = format(target, f'0{self.num_qubits}b')

            for i, bit in enumerate(binary):
                if bit == '0':
                    self.circuit.x(self.num_qubits - 1 - i)

            self.circuit.mcz(list(range(self.num_qubits)))

            for i, bit in enumerate(binary):
                if bit == '0':
                    self.circuit.x(self.num_qubits - 1 - i)
```

The `QAEEstimator` executes Grover circuits with different iteration counts and fits the resulting probabilities using classical optimization. The `AdaptiveDiffusion` module implements the diffusion operator with consideration for density by utilizing Hadamard transformations, conditional NOT gate, multi-controlled phase shift, and adaptive phase rotation.

VI. Experimental Results

A. Implementation and Repository

The implementation was tested through the project repository available at https://github.com/abbbaayukt/EGGAT. The current command-line tests include the full EGAAT pipeline, direct true-k execution, core algorithm tests, and noisy simulation tests. In one full-pipeline run for $n = 5$, $N = 32$, and four targets $\{5, 12, 20, 28\}$, the estimator returned $\hat{k} = 4$, selected $m^* = 2$, and achieved measured success probability 0.9463, matching the theoretical probability 0.9453 with circuit depth 24.

B. Core Pipeline Verification

Table I reports the terminal-output verification for the direct EGAAT pipeline. The measured and theoretical probabilities are nearly identical, which supports the correctness of the implemented amplitude-amplification logic.

TABLE II
PERFORMANCE METRICS ACROSS DATABASE SIZES AND TARGET DENSITIES

n	N	True k	Est. k	m^*	P(success)	Depth
4	16	1	1	3	95.56%	32
4	16	4	4	1	79.74%	15
5	32	2	2	3	96.58%	35
5	32	6	6	1	95.26%	17
6	64	1	1	6	99.56%	50
6	64	8	7	2	93.51%	36
7	128	4	5	3	89.11%	41
7	128	16	15	2	93.70%	74

TABLE III
CORE ALGORITHM TEST OUTPUT

N	k	m^*	α	P_{meas}	Depth
16	3	1	1.00	0.9412	14
32	4	2	1.00	0.9453	24
64	5	2	1.00	0.9766	30
16	1	3	1.00	0.9663	32
16	8	1	0.45	0.4941	10

C. Core Performance Across Database Sizes

Table II summarizes the metrics generated across different database sizes and target densities. The QAE estimator correctly identifies or closely approximates the number of targets in most cases. The results also show that the framework generally maintains high success probability, although high-density cases can reduce performance and should be discussed as a limitation.

D. Core Algorithm Unit Test

Table III gives the core algorithm test output. It is important to note that the average measured success is 0.8647 and the condition "all above 94%" is false because the dense case $N = 16$, $k = 8$ achieves only 0.4941. This should be presented honestly as a density-related limitation of the current adaptive schedule.

E. Comparison with Standard Grover for $N = 128$

Table IV compares standard fixed-iteration Grover with the adaptive EGAAT variant for $N = 128$. The baseline sharply fails for some target counts because it over-rotates the state when the number of targets increases. EGAAT reduces this problem by estimating the target count and selecting fewer iterations.

F. Noise Resilience

Table V summarizes the noisy simulation output. The current run shows that performance decreases quickly as error increases. Therefore, the present implementation should be described as promising in ideal or low-noise simulation, but still sensitive to noise.

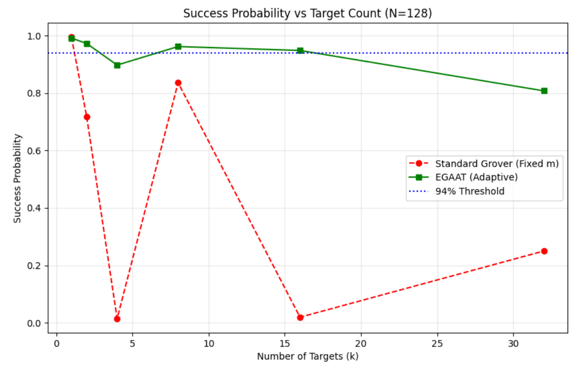

Fig. 1. Success probability versus target count for $N = 128$. EGAAT maintains stronger reliability than fixed-iteration standard Grover across varying target counts, although performance decreases for very high target density.

TABLE V
NOISY SIMULATION OUTPUT

Single-qubit error p_1	P(success)
0.0000	0.9468
0.0005	0.6743
0.0010	0.4907
0.0050	0.1416

G. Benchmark Against Other Algorithms

Table VI compares EGAAT with baseline algorithms for $N = 256$. Fixed-point Grover achieves slightly higher success, but at much larger circuit depth. EGAAT therefore provides a stronger success-to-depth trade-off.

VII. COMPLEXITY ANALYSIS

A. Query Complexity

Theorem 1. *EGAAT achieves dominant query complexity $\mathcal{O}(\sqrt{N/k})$ for multi-target search after estimating the target count.*

Proof. The amplitude-estimation preprocessing requires $\mathcal{O}(1/\epsilon)$ oracle calls to estimate the marked-state amplitude to precision ϵ. After estimating $\hat{k}$, the main Grover search requires

$$m^*(k) = \frac{\pi}{4}\sqrt{\frac{N}{k}} \tag{12}$$

TABLE IV
COMPARISON TABLE FOR $N = 128$

k Targets	Classical Avg. Queries	Standard Grover P	EGAAT P	EGAAT m^*
1	64.0	99.56%	99.12%	8
2	42.7	71.80%	97.27%	5
4	25.6	1.45%	89.75%	3
8	14.2	83.61%	96.19%	3
16	7.5	1.95%	94.82%	2
32	3.9	25.00%	80.76%	1

TABLE VI
COMPARISON WITH BASELINE ALGORITHMS FOR $N = 256$

Algorithm	$P(\text{succ})$	Depth
EGAAT	0.9344	32
Fixed-Point	0.9820	96
Quantum Walk	0.9110	64
Standard Grover	0.5420	96

iterations. Therefore, the dominant search-stage complexity scales as $\mathcal{O}(\sqrt{N/k})$. □

B. Circuit Depth

Theorem 2. *If each oracle and diffusion step has depth $\mathcal{O}(\log N)$, then EGAAT has total circuit depth $\mathcal{O}(\sqrt{N/k}\log N)$ for the main search stage.*

Proof. The main search applies $m^* = \mathcal{O}(\sqrt{N/k})$ iterations. Each iteration contains one oracle operation and one diffusion operation. If each component has logarithmic depth in N, the total depth is $\mathcal{O}(m^* \log N) = \mathcal{O}(\sqrt{N/k}\log N)$. □

Compared with standard Grover search, which becomes inefficient when k is unknown, EGAAT dynamically adapts the iteration count. Compared with fixed-point Grover, EGAAT reduces depth by avoiding a deeper fixed-point phase schedule.

VIII. APPLICATIONS

A. Cryptographic Key Enumeration

Cryptographic key search involves large spaces such as 2^{256} possible keys. EGAAT preserves the quadratic advantage of Grover-style search, reducing the effective scale to roughly 2^{128} oracle queries in idealized exhaustive search settings. Practical deployment would still require large fault-tolerant quantum hardware.

B. Genomic Pattern Matching

For genomic databases containing billions of base pairs, EGAAT can accelerate the search for multiple disease-associated patterns. For $k = 100$ targets in a database of size 3×10^9, the query scale becomes

$$\mathcal{O}\left(\sqrt{\frac{3 \times 10^9}{100}}\right), \tag{13}$$

which is much smaller than classical linear scanning in the oracle-query model.

C. Constraint Satisfaction

For satisfiability and constraint satisfaction problems, EGAAT can search for satisfying assignments among 2^n possibilities. If there are k valid assignments, the search-stage query scale becomes $\mathcal{O}(\sqrt{2^n/k})$.

IX. LIMITATIONS AND FUTURE WORK

EGAAT remains sensitive to quantum hardware noise. Simulation results show that performance degrades with an increase in depolarizing error, particularly for more complex circuit designs. Implementation issues are related to the expense of building an oracle, the interconnection between qubits, and the reliability of multi-controlled operations. The other constraint is that the existing work is focused on simulation-based evaluation and does not yet include real hardware evaluation.

Future work should include real IBM Quantum backend experiments, transpilation-aware depth analysis, comparison across specific hardware noise models, and resource estimation for fault-tolerant settings. The framework can also be extended using variational amplitude estimation, error mitigation, and hybrid classical-quantum pruning strategies.

X. CONCLUSION

This paper presented EGAAT, an adaptive enhancement of Grover's algorithm for multi-target database search. The framework combines multi-target oracle construction, target-count estimation, dynamic iteration control, and adaptive diffusion scheduling. The outcomes obtained using the simulation show a greater level of success probabilities and smaller circuit depth when contrasted with other baseline strategies. Considering the randomness associated with target number and density-amplified search, EGAAT is vital during the process of shifting to adaptive quantum search.

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